

1	Course Name:
	DATA STRUCTURES AND ALGORITHMS
2	Course Code:
	ISB30503
3	Year and Semester Offered:
	Year 2 Semester 1
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	. This course provides students with solid foundations in the basic concepts of efficient programming and problem-solving skills. It covers essential data structures and algorithms, with a focus on their applications in both IT and AI fields. The course includes topics such as lists, stacks, queues, trees, and graphs, with an emphasis on graph algorithms, including intelligent algorithms such as Depth First Search (DFS), Breadth First Search (BFS) and Dijkstra's algorithms.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Apply the concept of data structures for solving computing problems. 2. Evaluate the various types of data structures and their applications. 3. Build computer programs for implementing data structures using Java IDE.
8	Content Outline:
	Topic
	1. Introduction to Data Structures and Algorithms
	2. Algorithms Efficiency Analysis
	3. Java Collection Framework
	4. ArrayList & LinkedList
	5. Stack & Queue
	6. Binary Search Tree
	7. Balanced Binary Search Tree (AVL)
	8. Graph Terminologies, Applications and Representations
	9. Graph Traversal and Shortest Path Algorithms
9	References:

Nalluri Venkata Madhu Bindu, Programming, Data Structures and Algorithms: GATE-2024 [Data Science & Artificial Intelligence], LAP LAMBERT Academic Publishing, 2023
